

SYDNEY BOYS' HIGH SCHOOL

MOORE PARK, SURRY HILLS



Year 11 YEARLY EXAMINATIONS - September 2002

MATHEMATICS

Extension 1

Time allowed — Ninety Minutes
Examiner: A.M.Gainford

DIRECTIONS TO CANDIDATES

- *ALL* questions may be attempted.
- All necessary working should be shown in every question. Full marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used.
- Use a new booklet for each question.
- If required, additional paper may be obtained from the Examination Supervisor upon request.

Question 1. (18 Marks)

- (a) Simplify $\frac{x+1}{x}$ where $x = \sqrt{2} + 1$, expressing your answer with rational denominator. 2
- (b) Solve for x : 6
- (i) $|4 - x| = 3$
- (ii) $x^2 - 9 < 0$
- (iii) $\frac{1}{x-2} < 1$
- (c) Find the remainder when the polynomial $P(x) = x^3 - 4x$ is divided by $x+3$. 1
- (d) Simplify $\frac{x^3 + 27}{x^2 - 3x + 9}$ 2
- (e) Give the general solution of the equation $\cos\left(\theta + \frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}$. 2
- (f) Find the focus and directrix of the parabola $y = \frac{x^2}{8} + x + \frac{1}{4}$. 2
- (g) Express $\cos 4\theta$ as an expression in powers of $\cos \theta$ only. 3

Question 2. (18 Marks)

- (a) Differentiate: 8
- (i) $2x^3 - 4x + 1$
- (ii) $\sqrt{x} - \frac{1}{x}$
- (iii) $(x^2 - 2)^5(2x + 1)$
- (iv) $\frac{4}{4 - x^2}$
- (b) (i) Express $\sqrt{2} \sin \theta + \sqrt{2} \cos \theta$ in the form $R \sin(\theta + \alpha)$, where $R > 0$, α is acute. 4
- (ii) Hence or otherwise sketch the graph of $y = \sqrt{2} \sin \theta + \sqrt{2} \cos \theta$ in the domain $0 \leq \theta \leq 2\pi$.

- (c) Find the exact value of $\operatorname{cosec} 105^\circ$. 1
- (d) How many terms of the arithmetic series $96 + 93 + 90 + \dots$ must be taken to give a sum of zero? 2
- (e) Given the polynomial $P(x) = x^3 + 3x^2 - x - 3$. 3
- (i) Use the factor theorem to find a zero of the polynomial.
- (ii) Express $P(x)$ as a product of three linear factors.

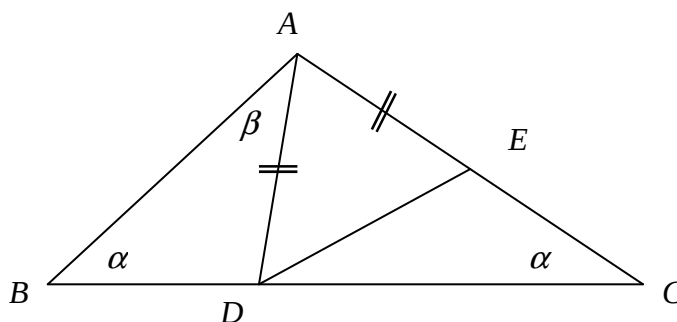
Question 3. (18 Marks)

- (a) Given $\log_a 5 = 0.827$ and $\log_a 2 = 0.356$, find $\log_a 50$. 2
- (b) Draw neat sketches of the following functions, showing their principle features: 6
- (i) $y = 1 - |x|$ (ii) $y = \log_2 x$ (iii) $y = \sqrt[3]{x}$
- (c) Given the function $f(x) = 2^x + 2^{-x}$ 7
- (i) Find $f(-1)$.
- (ii) Show that $f(x)$ is even.
- (iii) Find $f(x) = 0$.
- (iv) State the domain and range of $f(x)$.
- (v) Sketch the function.
- (d) A geometric series has 4 as its third term and $-\frac{32}{27}$ as its sixth term. Find the first term and the common ratio. 2
- (e) Calculate to the nearest minute the acute angle between the lines $5x - 4y = 17$ and $3x + 2y = 8$. 1

Question 4. (18 Marks)

(a)

4



In the isosceles triangle ABC , $\angle ABC = \angle ACB = \alpha$. The points D and E lie on BC and AC , so that $AD = AE$, as shown in the diagram. Let $\angle BAD = \beta$.

- (i) Explain why $\angle ADC = \alpha + \beta$.
- (ii) Find $\angle DAC$ in terms of α and β .
- (iii) Hence, or otherwise, find $\angle EDC$ in terms of β .

4

- (b) Let $P(2ap, ap^2)$ be a point on the parabola $x^2 = 4ay$.

- (i) Write down the equation of the tangent at P .
- (ii) Find the co-ordinates of T , the point where the tangent meets the axis of the parabola.
- (iii) Show that the tangent at P is equally inclined to the axis of the parabola and to the line joining P to the focus S .

- (c) Find the value of the constants a and b if $x^2 - 2x - 3$ is a factor of the polynomial $P(x) = x^3 - 3x^2 + ax + b$ 2

- (d) The point $P(11, 7)$ divides AB externally in ratio 3:1. If B is $(6, 5)$, find the co-ordinates of A . 2

- (e) An observer (O) in a lighthouse is 180m vertically above a point B at sea level on the shore. He observes the frigate HMS Shropshire (S) on a bearing of $210^\circ T$ and an angle of depression of $5^\circ 21'$. He also notes that Wolf Rock (W) is on a bearing of $165^\circ T$ and an angle of depression of $3^\circ 40'$. 6

- (i) Sketch a diagram to represent this situation.
- (ii) Calculate the distance of the frigate from the rock, to the nearest metre.
- (iii) Calculate the true bearing from the frigate to the rock.

This is the end of the paper.



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SEPTEMBER 2002

YEARLY EXAMINATION

YEAR 11

Mathematics Extension

Sample Solutions

Question (1) [18 marks]

(a) $\frac{x+1}{x} = 1 + \frac{1}{x}$

$= 1 + \frac{1}{\sqrt{2}+1}$

$= 1 + \frac{1}{\sqrt{2}+1} \times \frac{\sqrt{2}-1}{\sqrt{2}-1}$

$= 1 + (\sqrt{2}-1)$

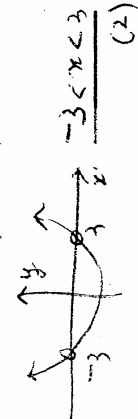
$= \boxed{\sqrt{2}}$

(b) $|4-x| = 3$

(i) $4-x = \pm 3$

$x = 1$ or $x = 7$ (2)

(ii) $(x+3)(x-3) < 0$



(iii) $\frac{1}{x-2} < 1$ (2)

Boundary: $x = 2$

$\frac{1}{x-2} = 1$

$x-2 = 1$

$x = 3$



$x < 2$ or $x > 3$ (2)

(c) $p(x) = x^3 - 4x$

$p(-3) = -27 + 12 = -15$

(d) $\frac{x^3+27}{x^2-3x+9}$ (1)

$= \frac{(x+3)(x^2-3x+9)}{(x^2-3x+9)} = x+3$ (2)

(e) $\cos(\theta + \frac{\pi}{4}) = \cos \frac{\pi}{4}$

$\theta + \frac{\pi}{4} = 2n\pi \pm \frac{\pi}{4}$

$\theta = 2n\pi \pm \frac{\pi}{4} - \frac{\pi}{4}$ (2)

$\theta = \begin{cases} 2n\pi \\ 2n\pi - \frac{\pi}{2} \end{cases}$

(f) $y = \frac{x^2}{8} + x + \frac{1}{x}$

$8y = x^2 + 8x + 2$

$= (x^2 + 8x + 16) - 14$

$\therefore (x+4)^2 = 8y + 14$

$(x+4)^2 = 8(y + \frac{7}{4})$

$4a = 8, a = 2$

vertex $(-4, -\frac{7}{4})$

focus $(-4, \frac{1}{4})$ (2)

directrix $y = -3\frac{3}{4} (-\frac{15}{4})$

(g) $\cos 4\theta = 2\cos^2\theta - 1$

$= 2(\cos^2\theta) - 1$

$= 2[2\cos^2\theta - 1] - 1$

$= 2[4\cos^4\theta - 4\cos^2\theta + 1] - 1$

$= 8\cos^4\theta - 8\cos^2\theta + 1$ (1)

(3)

Q2 a, (i) $\frac{d}{dx} (2x^3 - 4x + 1)$

$$= 6x^2 - 4$$

1

(ii) $\frac{d}{dx} \left(\sqrt{x} - \frac{1}{x} \right)$

$$= \frac{1}{2} x^{-\frac{1}{2}} + \frac{1}{x^2}$$

2

(iii) $\frac{d}{dx} (x^2 - 2)^5 (2x + 1)$

$$= (2x + 1) \cdot 10x (x^2 - 2)^4 + (x^2 - 2)^5 \cdot 2$$

$$= 2(x^2 - 2)^4 (5x(2x + 1) + x^2 - 2)$$

$$= 2(x^2 - 2)^4 (10x^2 + 5x + x^2 - 2)$$

$$= 2(x^2 - 2)^4 (11x^2 + 5x - 2)$$

$$u = (x^2 - 2)^5$$

$$u' = 5(x^2 - 2)^4 \cdot 2x$$

$$= 10x(x^2 - 2)^4$$

$$v = 2x + 1$$

$$v' = 2$$

3

(iv) $\frac{d}{dx} \left(\frac{4}{4 - x^2} \right)$

$$= \frac{d}{dx} \left(4(4 - x^2)^{-1} \right)$$

$$= 4 \cdot -1 (4 - x^2)^{-2} \cdot -2x$$

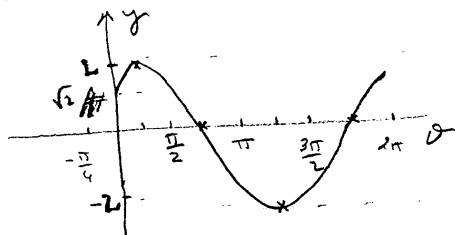
$$= \frac{8x}{(4 - x^2)^2}$$

2

$$\begin{aligned}
 (b) \quad & \sqrt{2} \sin \theta + \sqrt{2} \cos \theta \\
 &= 2 \left(\frac{1}{\sqrt{2}} \sin \theta + \frac{1}{\sqrt{2}} \cos \theta \right) \\
 &= 2 \left(\sin \theta \cos \frac{\pi}{4} + \cos \theta \sin \frac{\pi}{4} \right) \\
 &= 2 \sin \left(\theta + \frac{\pi}{4} \right)
 \end{aligned}$$

2

(ii)



$\sin \theta$
Zeros at $\theta = 0, \pi, 2\pi$

$\sin \left(\theta + \frac{\pi}{4} \right)$
Zeros at $\theta + \frac{\pi}{4} = 0, \pi, 2\pi$
 $\theta = -\frac{\pi}{4}, \frac{3\pi}{4}, \frac{7\pi}{4}$

2

(c) $\cos 105^\circ$

$$\begin{aligned}
 &= \frac{1}{\sin(60+45)^\circ} \\
 &= \frac{1}{\sin 60^\circ \cos 45^\circ + \cos 60^\circ \sin 45^\circ} \\
 &= \frac{1}{\frac{\sqrt{3}}{2} \times \frac{1}{\sqrt{2}} + \frac{1}{2} \times \frac{1}{\sqrt{2}}} \\
 &= \frac{2\sqrt{2}}{\sqrt{3} + 1}
 \end{aligned}$$

1

$$(d) \quad S_n = \frac{n}{2} (2a + (n-1)d)$$

$$0 = \frac{n}{2} (192 + (n-1) \times -3)$$

$$\therefore 0 = n(192 - 3n + 3)$$

$$0 = n(195 - 3n)$$

$$\therefore n = 0 \quad \text{or} \quad 3n = 195$$

$$\therefore n = 65$$

$\therefore 65$ terms required

2

$$(2) \quad P(x) = x^3 + 3x^2 - x - 3$$

$$(i) \quad P(1) = 1 + 3 - 1 - 3 = 0$$

$\therefore x-1$ is a factor.

1

$$(ii) \quad P(x) = \cancel{x^3 + 3x^2} x^2(x+3) - 1(x+3)$$

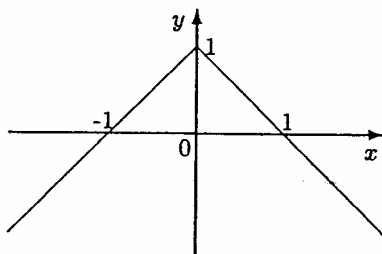
$$= (x+3)(x^2-1)$$

$$= (x+3)(x+1)(x-1)$$

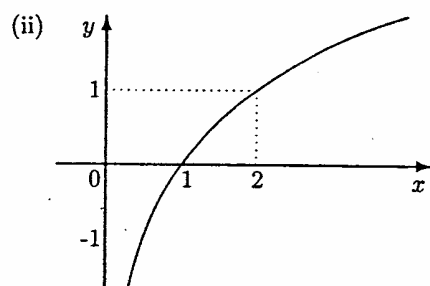
2

2 3. (a) $\log_a 50 = \log_a 5^2 \times 2,$
 $= 2 \log_a 5 + \log_a 2,$
 $= 2 \times 0.827 + 0.356,$
 $= 2.01.$

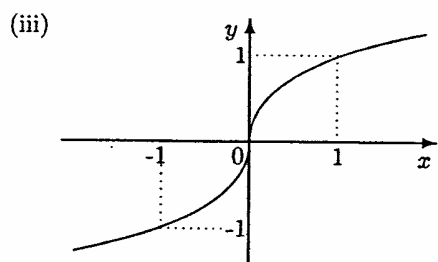
2 (b) (i)



2



2



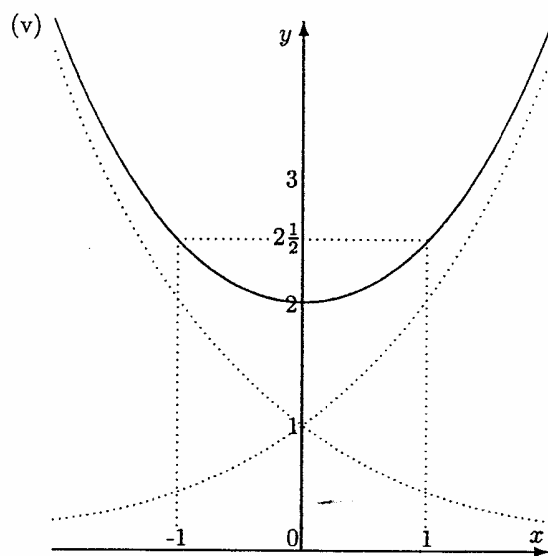
1 (c) (i) $f(-1) = \frac{1}{2} + 2,$
 $= 2\frac{1}{2}.$

1 (ii) $f(-x) = 2^{-x} + 2^{-(-x)},$
 $= 2^{-x} + 2^x,$
 $= f(x).$
 $\therefore$ the function is even.

1 (iii) $2^x > 0$ and $2^{-x} > 0,$
 $\therefore 2^x + 2^{-x} \neq 0,$ i.e., there is no solution.

2 (iv) Domain is all real $x.$
Range is $f(x) \geq 2.$

2



2

(d) With the usual notation:

$$ar^2 = 4,$$

$$ar^5 = \frac{-32}{27}.$$

$$\text{Now, } \frac{ar^5}{ar^2} = \frac{-32}{27} \times \frac{1}{4},$$

$$= \frac{-8}{27},$$

$$= r^3.$$

$$\text{So } r = -\frac{2}{3}.$$

$$a \times \left(-\frac{2}{3}\right)^2 = 4,$$

$$\text{Hence } a = 4 \times \frac{9}{4},$$

$$= 9.$$

1

(e) $m_1 = \frac{5}{4}$ and $m_2 = -\frac{3}{2}$,

$$\tan^{-1} \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| = \tan^{-1} \left(\frac{22}{7} \right),$$

$$\approx 72^\circ 21' \text{ (Nearest minute).}$$

Alternative method:

$$\tan^{-1} \left(\frac{5}{4} \right) \approx 51^\circ 20' 25'',$$

$$\tan^{-1} \left(-\frac{3}{2} \right) \approx 123^\circ 41' 24'',$$

$$123^\circ 41' 24'' - 51^\circ 20' 25'' \approx 72^\circ 21' \text{ (Also to the nearest minute).}$$

Question 4

(a) (i) $\angle ADC = \alpha + \beta$ (exterior angle is equal to the sum of the internal opposite angles.)

(1)

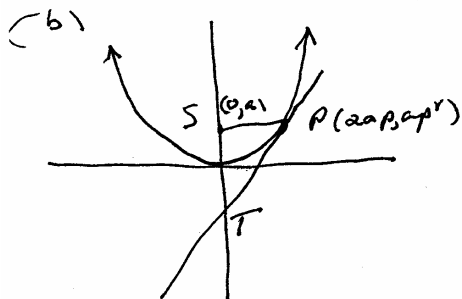
(ii) $\angle DAC = 180 - \alpha - \alpha - \beta$
 $= 180 - 2\alpha - \beta$

(1)

(iii) $\therefore \angle ADE = \frac{2\alpha + \beta}{2}$ base angle of isosceles Δ .
 $= \alpha + \frac{\beta}{2}$

$\therefore \angle EDC = \angle ADC - \angle ADE$
 $= \alpha + \beta - (\alpha + \frac{\beta}{2})$
 $= \frac{\beta}{2}$

(2)



(i) $y - tx + at^2 = 0$

(1)

(ii) T is $(0, -at^2)$

(1)

(iii) $SP = \sqrt{(ap^2 - a)^2 + (2ap - 0)^2}$
 $= \sqrt{a^2 p^4 - 2a^2 p^2 + a^2 + 4a^2 p^2}$
 $= \sqrt{a^2 (p^4 + 2p^2 + 1)}$
 $= a(p^2 + 1)$

$ST = a + ap^2$
 $= a(p^2 + 1)$

(2)

$\therefore ST = SP$

$\therefore \Delta PST$ is isosceles

$\therefore \angle PTS = \angle SPT$

(c) If $x^2 - 2x - 3 = (x-3)(x+1)$

is a factor of $P(x) = x^3 - 3x^2 + ax + b$.

then $x-3$ is a factor $\Rightarrow P(3) = 0$

& $x+1$ is a factor $\Rightarrow P(-1) = 0$

} alle.
①

Here: $27 - 27 + 3a + b = 0$

$3a + b = 0$ — ①

and $-1 - 3 - a + b = 0$

$-a + b = 4$ — ②

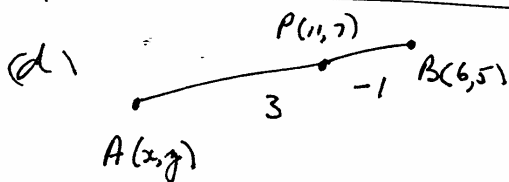
① - ②

$4a = -4$

$\boxed{a = -1}$

And in ① $-3 + b = 0$

$\boxed{b = 3}$



$11 = \frac{3 \times 6 + (-1) \times x}{2}$

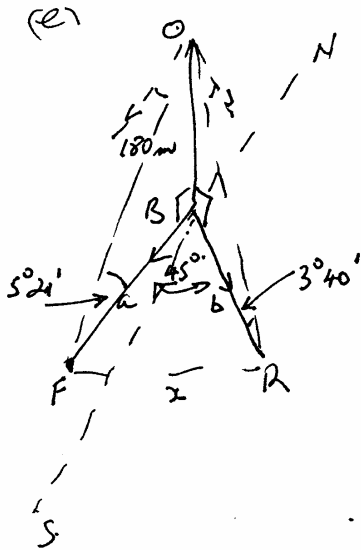
$22 = 18 - x$

$\boxed{x = -4}$

$7 = \frac{3 \times 5 + (-1) \times y}{2}$

$14 = 15 - y$

$\boxed{y = 1}$ ∴ Pt (-4, 1).



(i) Sketch.

①

$$(ii) x^2 = a^2 + b^2 - 2ab \cos 45^\circ.$$

$$\text{where } a = 180 \tan 84^\circ 39'$$

$$\text{and } b = 180 \tan 86^\circ 20'.$$

$$\therefore x^2 = 180^2 \tan^2 84^\circ 39' + 180^2 \tan^2 86^\circ 20' - 2 \times 180^2 \tan 84^\circ 39' \tan 86^\circ 20' \times \frac{1}{\sqrt{2}}.$$

$$x^2 = 180^2 \left[\tan^2 84^\circ 39' + \tan^2 86^\circ 20' - \sqrt{2} \tan 84^\circ 39' \times \tan 86^\circ 20' \right]$$

$$x = 180 \sqrt{(\quad)} \quad \text{③}$$

$$\approx 1987 \text{ m}$$

$$(iii) \frac{\sin \angle BFR}{180 \tan 86^\circ 20'} = \frac{\sin 45^\circ}{1987}$$

$$\sin \angle BFR = \frac{180 \tan 86^\circ 20' \sin 45^\circ}{1987}$$

$$= 0.99957 \dots$$

$$\angle BFR = \sin^{-1}(\quad)$$

$$= 88^\circ 20'.$$

$\therefore$ Bearing of R from S is $118^\circ 20' \text{ T}$

②